

Wednesday Warm Up: Unit 1, Capacitors, Current and DC circuits

Prof. Jordan C. Hanson

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1 Memory Bank

- $\Delta U = q\Delta V$... Relationship between change in potential energy, charge, and voltage.
- $V(r) = kq/r$... Potential (voltage) of a point charge.
- $V(x) = Ex + V_0$... Linear relationship between voltage, electric field, and distance between two charge plates to form a capacitor.
- $Q = C\Delta V$... Relationship between stored charge, capacitance, and voltage. Capacitance has the unit of 1 Farad, or 1 C V⁻¹.
- $I = \Delta Q/\Delta t$... The relationship between current, I , and charge. Current has a unit of 1 Amp (A), or 1 C s⁻¹.
- $q_e = 1.60 \times 10^{-19}$ C ... The charge of 1 electron.

2 Review: Electric Fields and Charge

1. Consider a particle of charge q and mass m passing through the region between a pair of infinitely long horizontal plates separated by a distance d with a uniform electric field strength E in the $+\hat{i}$ direction. At $t = 0$, the particle is at the origin, half-way between the plates. (a) Create an expression for the horizontal displacement of the particle, $\Delta x(t)$. (b) Let $E = 500$ N C⁻¹, $m = 7 \times 10^{-27}$ kg, $q = 3.2 \times 10^{-19}$ C, $d = 3$ cm, and $v_i = 0.1$ m s⁻¹ $\hat{j}$.

3 Charge, Voltage, and Potential Energy

1. Suppose a 1 nC charge is situated at the origin. (a) Calculate the voltage felt by a test charge 1.2 meters from the origin. (b) What is the voltage 2.4 m from the origin?
2. Suppose the capacitor in the first exercise shoots out a 1 nC charge every 100 ms. (a) To what current does this correspond? (b) How many electrons per second is this?

4 Capacitors and Current

1. (a) Suppose we allow 5 mA of current to flow onto a capacitor for 5 seconds. How much charge is stored in the capacitor? (b) After 5 seconds, the capacitor voltage is measured to be 5.0 V. What is the capacitance?